

Real Estate Flow — addToTable / updateTableRow Design

Source: op1Screen rows 704–836+ (Vertika Tekno Lokacipta real estate pages)

Reference page: vertikaTeknoLokaciptaReportMeetingOwner (row 704–721)

1. addToTable DSL — token cheatsheet

■key■value■key■value■<N>■value...

Symbol	Codepoint	Meaning
■	U+2B58	Pair separator (between key/value pairs)
■	U+25FC	Inner separator (key from value)
<N>	literal	Target table column index (1-based, positional)
■N■	U+25C1 / U+25B7	Reference to widget value at op1Screen seq# N within current page (col A)
■ ... ■	U+25C0 / U+25B6	Formatter token (e.g. `2\
literal	—	Static value baked into formula

Header keys (always at start, no <N> slot):

- retention — TTL in minutes (4320 = 3 days)
- description — activity domain (report-activity, report-prospect)
- flag — activity subtype label (meeting, marketing, etc.)

Body = numbered slots <1> to <N> mapped to target table columns.

2. Visual walkthrough — Meeting Owner page (row 704)

Page widget rows (op1Screen!A705–A720)

seq#	widget	label	input type
1	topMain	(header gradient)	static
2	separator	-	-
3	text	"Meeting Owner"	static title
4	textField	Nama Klien	text
5	textField	Property	text
6	lqrTextField1	(Property QR scan)	qr-link → master
7	numericField	Asking Price	number
8	separator	-	-
9	displayDropDownData	Tipe Properti	enum (House/Apartment/Land/Ruko/...)
10	displayDropDownData	Lead Source	enum (Canvassing/Referral/Online/...)
11	separator	-	-
12	getImage1	Property Photo	photo[]
13	radio	Hasil Aktivitas	radio (Potential/Follow Up/...)
14	3LineBorderForm	Catatan	textarea
15	displayDropDownData	Next Action	enum (Follow-up/Showing/Proposal/...)
16	sendButtonGpsAddTable	"Kirim"	submit (carries addToTable)

addToTable string at row 720 col D (the Kirim button)

■retention■4320

```

■description■report-activity
■flag■meeting
■<1>■report-prospect-owner
■<2>■2|T7|Ddd MMM yyyy HH:mm■
■<3>■87544551624342
■<4>■Agenia Demo-7
■<5>■5■
■<6>■11■
■<7>■10■
■<8>■Product Group
■<9>■Product Group
■<10>■3■
■<11>■4■
■<12>■12■
■<13>■13■
■<14>■4■

```

How frontend resolves it on submit

TARGET TABLE ROW (positional):	
col 1 = "report-prospect-owner"	[literal]
col 2 = "11 May 2026 14:32"	[server time, formatted]
col 3 = "87544551624342"	[employee VID, baked]
col 4 = "Agenia Demo-7"	[employee name, baked]
col 5 = <value of widget seq#5>	[Property text – "Villa Permata"]
col 6 = <value of widget seq#11>	[separator – empty? bug or padding]
col 7 = <value of widget seq#10>	[Lead Source – "Referral"]
col 8 = "Product Group"	[cost center, baked]
col 9 = "Product Group"	[cost center 2, baked]
col 10 = <value of widget seq#3>	[Title text "Meeting Owner" – header]
col 11 = <value of widget seq#4>	[Nama Klien – "Pak Budi"]
col 12 = <value of widget seq#12>	[Property Photo URLs JSON]
col 13 = <value of widget seq#13>	[Hasil Aktivitas – "Follow Up"]
col 14 = <value of widget seq#4>	[Nama Klien again – duplicate intentional]

Plus auto-injected by sendButtonGpsAddTable

- `gpsPosition: 2` → adds GPS lat/lng as col 2 input (parallel column or merged)
- `chain: DO_DIALOG` → after save, show "Terkirim" + Ok button → route back

3. Why slot mapping looks weird

Looking at `<5>■5■ <6>■11■ <7>■10■ <10>■3■ <11>■4■` — slots not in widget row order. This is because **target table column order differs from form display order**. Designer arranged form by UX flow (Client first, then Property), but DB table columns ordered by reporting needs (Property first, then Client metadata).

This is the **core design decision**: separate UX layout from DB schema via positional mapping.

4. All 8 real estate pages — full schema

Page A: Meeting Owner (704–721)

- Title: Meeting Owner
- Inputs: Nama Klien, Property name, Property QR, Asking Price, Tipe Properti, Lead Source, Property Photo, Hasil Aktivitas, Catatan, Next Action
- Submit flag: `report-prospect-owner`

Page B: Add Listing (722–741)

- Title: Add Listing
- Inputs: Nama Klien, Property name, Property QR, Tipe Properti, **Listing Type** (radio: Sale/Rent), Asking Price, **Negotiable** (checkbox), Lead Source, Property Photo, Hasil Aktivitas, Catatan, Next Action
- Submit flag: `report-prospect-owner` (likely should be `report-listing-add`)

Page C: Marketing Property (742–761)

- Title: Marketing Property
- Inputs: Nama Klien, Property name, Property QR, Tipe Properti, **Marketing Activity** (radio), Asking Price, Negotiable, Lead Source, Property Photo, Hasil Aktivitas, Catatan, Next Action
- Submit flag: `report-prospect-owner` (should be `report-marketing-property`)

Page D: Meeting Buyer (762–774)

- Title: Meeting Buyer
- Inputs: Property QR, **Hasil Aktivitas** (DRD: Potential/Follow Up/Not Interested), Nama Klien, Next Action, Property Photo, Catatan
- Submit flag: `report-activity-meeting`

Page E: Showing (775–787)

- Title: Showing
- Inputs: Nama Buyer, Property QR, **Hasil Jenis Interest**, Next Action, Property Photo, Catatan
- Submit flag: `report-activity-showing`

Page F: Follow Up (788–802)

- Title: Showing (misabeled — should be Follow Up)
- Inputs: Nama Buyer, Property QR, **Channel** (WhatsApp/Telp/Email), **Conversation Result**, Next Action, Property Photo, Catatan
- Submit flag: `report-activity-followup`

Page G: Deal Progress (803–819)

- Title: Showing (misabeled — should be Deal Progress)
- Inputs: Nama Buyer, Property QR, **Deal Stage** (DRD), **Price** (numeric), **Expected Closing** (datePicker), Next Action, Property Photo, Catatan
- Submit flag: `report-activity-deal`

Page H: Prospect Owner (820–836+)

- Title: Prospect Owner
- Inputs: Location (text), Property QR, Tipe Properti, Lead Source, Nama Klien, Property Photo, Hasil Aktivitas, **Keterangan**, Next Action
- Submit flag: `report-prospect-*`

5. Data dictionary — union of all real estate fields

Auto / system (no widget, baked into addToTable formula)

Field	Source	Example
myVid	session VID	87544551624342
employeeName	session profile	Agenia Demo-7
timestamp	server time formatter `■2\`	T7\
costCenter1	session org	Product Group
costCenter2	session org parent	Product Group
gpsLatLng	sendButtonGpsAddTable auto	[-6.31607,106.64483]
activityDomain	header description	report-activity / report-prospect
activitySubtype	header flag	meeting, marketing, ...
reportTag	<1> literal	report-prospect-owner, report-activity-meeting

Client side

Field	Widget	Type	Pages
namaKlien	textField	text	A, B, C, H
namaBuyer	textField	text	D, E, F, G
channel	DRD	enum: WhatsApp, Telp, Email, Direct	F

Property side

Field	Widget	Type	Pages
propertyName	textField	text	A, B, C
propertyCode	lqrTextField1	qr-link → property master	A–H
location	text	text (free)	H
tipeProperti	DRD	enum: House, Apartment, Land, Ruko, Office, Villa	A, B, C, H
listingType	radio	enum: Sale, Rent	B
propertyPhoto	getImage1	string[] (URLs)	A–H

Commercial

Field	Widget	Type	Pages
askingPrice	numericField	money	A, B, C
price	numericField	money (final/agreed)	G
negotiable	checkBox	bool	B, C
leadSource	DRD	enum: Canvassing, Referral, Online, Walk-in, Cold-call, Social Media	A, B, C, H

Activity outcome

Field	Widget	Type	Pages
hasilAktivitas	radio / DRD	enum: Potential, Follow Up, Not Interested, Closed	A, B, C, D, H
hasilJenisInterest	DRD	enum: High, Medium, Low	E
marketingActivity	radio	enum: Brochure, Open House, Tour, Ad	C
conversationResult	DRD	enum: Reached, Voicemail, No Answer	F
dealStage	DRD	enum: Initial, Negotiation, Offer Sent, Closed Won, Closed Lost	G

Field	Widget	Type	Pages
expectedClosing	datePicker	date	G

Continuation

Field	Widget	Type	Pages
nextAction	DRD	enum: Follow-up, Showing, Proposal, Close, Drop	A–H
catatan	3LineBorderForm	textarea	A–G
keterangan	3LineBorderForm	textarea	H

6. Recommended target table — `crm_realestate_activity`

Single positional table, discriminator-based.

col	name	type	source	notes
1	activity_subtype	string	<1> literal	report-* tag
2	timestamp	datetime	<2> formatter ■...■	server time
3	employee_vid	string	<3> session	FK → employee
4	employee_name	string	<4> session	denormalized
5	property_code	string	<5> ■lqr■	FK → property master
6	property_name	string	<6> ■text■	freeform display
7	client_name	string	<7> ■text■	klien OR buyer
8	cost_center_code	string	<8> session	
9	cost_center_name	string	<9> session	
10	tipe_properti	enum	<10> ■DRD■	nullable
11	lead_source	enum	<11> ■DRD■	nullable
12	asking_price	number	<12> ■numeric■	nullable
13	hasil_aktivitas	enum	<13> ■radio/DRD■	nullable
14	next_action	enum	<14> ■DRD■	nullable
15	property_photo	json[]	<15> ■getImages■	URLs
16	notes	string	<16> ■textarea■	catatan/keterangan
17	gps_lat_lng	json	gpsPosition auto	[lat, lng]
18	listing_type	enum?	<18> ■radio■	only B
19	negotiable	bool?	<19> ■checkBox■	only B, C
20	marketing_activity	enum?	<20> ■radio■	only C
21	channel	enum?	<21> ■DRD■	only F
22	conversation_result	enum?	<22> ■DRD■	only F
23	deal_stage	enum?	<23> ■DRD■	only G
24	expected_closing	date?	<24> ■datePicker■	only G
25	price_final	number?	<25> ■numeric■	only G
26	location_text	string?	<26> ■text■	only H

26 cols total. Most pages fill cols 1–17, page-specific extras fill 18–26.

7. addToTable formula examples (paste into op1Screen)

Meeting Owner (Page A) — properly normalized

Assume widgets at seq#: 4=Nama Klien, 5=Property, 6=PropQR, 7=AskingPrice, 9=Tipe, 10=LeadSource, 12=Photo, 13=Hasil, 14=Catatan, 15=NextAction.

```

■retention■4320
■description■report-activity
■flag■meeting-owner
■<1>■report-meeting-owner
■<2>■2|T7|Ddd MMM yyyy HH:mm■
■<3>■3|VID■
■<4>■4|NAME■
■<5>■6■
■<6>■5■

```

```

■<7>■4■
■<8>■5|CC1■
■<9>■6|CC2■
■<10>■9■
■<11>■10■
■<12>■7■
■<13>■13■
■<14>■15■
■<15>■12■
■<16>■14■

```

Add Listing (Page B) — adds cols 18, 19

Append after col 17:

```

■<18>■<listingTypeRow>■
■<19>■<negotiableRow>■

```

Deal Progress (Page G) — adds cols 23, 24, 25

Append:

```

■<23>■<dealStageRow>■
■<24>■<expClosingRow>■
■<25>■<priceRow>■

```

8. updateTableRow design (from existing pattern)

Existing pattern (page-detail-full.json lines 103, 130):

```

$test/request-approval//vtl.trial-approval
■tablevid■20342033315492
■search■1★<request_vid>
■<19>■APPROVED
■<20>■1■
■<21>■2■
■<27>■1|T7|yyyy-MM-dd HH:mm:ss■

```

Header pairs:

- tablevid — Firebase node ID untuk table itu
- search — colN★value (★ = U+2605) cari row yg col N = value

Body: hanya kolom yg mau diupdate. Slot lain skipped (sparse patch).

Untuk Real Estate, primary key = request_vid / transaction_id di col 1, generated via NUMBER widget (lihat section 11).

Update pattern jadi:

```

$<env>/<workspace>//vtl.realestate-activity
■tablevid■<RE_TABLE_VID>
■search■1★<activity_id>
■<13>■<new hasilAktivitas>
■<14>■<new nextAction>
■<27>■1|T7|yyyy-MM-dd HH:mm:ss■

```

Sparse: hanya 3 kolom dipatch (status, nextAction, updated_at).

9. Bugs / inconsistencies spotted (worth fixing before adding more pages)

1. **Pages B, C, H all use flag** report-prospect-owner in addToTable header even though intent differs. Real differentiator is <1> slot. Header flag should mirror page intent (listing-add, marketing, prospect-owner).
2. **Pages F, G mislabeled "Showing"** in title text widget — copy-paste from Page E. Should be "Follow Up" / "Deal Progress".
3. **Static** 87544551624342 **and** Agenia Demo-7 **baked into 8 formulas** — single mock employee. Should template ■3|VID■ / ■4|NAME■ so formula works for any signed-in user. Same for Product Group cost center.
4. **Slot reuse**: row 720 has <11>■■4■ AND <14>■■4■ — Nama Klien written to two columns. Either intentional denorm or paste error. Verify.
5. **Slot 6 references seq#11 (separator)** — separator widgets have no value, will produce empty string. Likely off-by-one error.

10. Quick sanity check before implementation

Open vertikaTeknoLokaciptaReportMeetingOwner in app, fill the form, check what server actually receives. Compare against col-by-col map above. Adjust slot indices if any drift.

11. Unique transaction ID — NUMBER widget pattern

Reference: request-leave.json line 47–56.

```
{
  "type": "NUMBER",
  "text": "No. Permohonan:◆Akan di buat otomatis",
  "template": "REQ-{{YYYY}}-{{COUNTER(vtl.request,6)})",
  "textColor": "blue_700",
  "size": 18,
  "executable": "execute1,generate_number",
  "position": 17,
  "margin": "8,16,8,16"
}
```

How it works:

- template = format pattern. {{YYYY}} → tahun, {{COUNTER(vtl.request,6)}} → counter Firebase node vtl.request 6-digit zero-padded.
- executable: "execute1,generate_number" → triggered saat submit (run: "...17:generate_number..." di Kirim button).
- position: 17 → nilai disimpan di slot 17 widget state.
- Di addToTable, referenced sebagai <1>■■17■ → kolom 1 table = generated ID.

Output example: REQ-2026-000001, REQ-2026-000002, ...

Untuk Real Estate, recommended templates:

Page	Template	Counter node
Add Listing	RE-LST-{{YYYY}}-{{COUNTER(vtl.realestate-property,6)}}	property master
Meeting Owner	RE-MTG-{{YYYY}}-{{COUNTER(vtl.realestate-activity,6)}}	activity log
Meeting Buyer	RE-MTG-{{YYYY}}-{{COUNTER(vtl.realestate-activity,6)}}	(sama, beda subtype)
Showing	RE-SHW-{{YYYY}}-{{COUNTER(vtl.realestate-activity,6)}}	activity log

Page	Template	Counter node
Follow Up	RE-FUP-{{YYYY}}-{{COUNTER(vtl.realestate-activity,6)}}	activity log
Deal Progress	RE-DEAL-{{YYYY}}-{{COUNTER(vtl.realestate-deal,6)}}	deal pipeline
Prospect Owner	RE-PRS-{{YYYY}}-{{COUNTER(vtl.realestate-activity,6)}}	activity log

12. Recommended Firebase schema — 3 tables

Berdasarkan lifecycle entity bukan per-form:

Table 1: `vtl.realestate-property` (master, slow-update)

Diisi dari **Add Listing** page. Updated jarang (price change, status change).

col	name	type	source	indexed
1	property_id	string	NUMBER widget	YES (PK)
2	created_at	datetime	server	YES
3	created_by_vid	string	session	YES
4	created_by_name	string	session	
5	property_name	string	textField	YES (search)
6	property_code_qr	string	lqrTextField1	YES
7	location	string	textField	YES
8	tipe_properti	enum	DRD	YES (filter)
9	listing_type	enum	radio: Sale/Rent	YES
10	asking_price	number	numericField	YES (sort)
11	negotiable	bool	checkBox	
12	photos	json[]	getImages1	
13	gps	json	gpsPosition	
14	status	enum	(Available/Sold/Hold)	YES
15	updated_at	datetime	server	

Table 2: `vtl.realestate-activity` (log, append-heavy)

Diisi dari **Meeting Owner, Meeting Buyer, Showing, Follow Up, Marketing, Prospect Owner**. High volume, immutable rows (insert-only). Update only status outcome.

col	name	type	source	indexed
1	activity_id	string	NUMBER widget	YES (PK)
2	created_at	datetime	server	YES
3	activity_subtype	enum	(meeting-owner, meeting-buyer, showing, follow-up, marketing, prospect)	YES (f...)
4	property_id	string	FK → property	YES (join)
5	property_name	string	denorm display	
6	property_code_qr	string	lqr scan	
7	client_name	string	textField	YES (search)
8	client_role	enum	(owner/buyer)	
9	employee_vid	string	session	YES
10	employee_name	string	session	
11	cost_center_vid	string	session	
12	cost_center_name	string	session	
13	hasil_aktivitas	enum	radio/DRD	YES
14	next_action	enum	DRD	
15	photos	json[]	getImages1	
16	notes	string	3LineBorderForm	
17	gps	json	gpsPosition auto	
18	channel	enum?	only Follow Up	
19	conversation_result	enum?	only Follow Up	
20	marketing_activity	enum?	only Marketing	
21	tipe_properti	enum?	denorm from prop	
22	lead_source	enum?	only Prospect/Add	
23	status	enum	(DONE/PENDING)	YES
24	updated_at	datetime		

Table 3: `vtl.realestate-deal` (pipeline, state machine)

Diisi dari **Deal Progress** page. Updated saat stage berubah.

col	name	type	source	indexed
1	deal_id	string	NUMBER widget	YES (PK)
2	created_at	datetime	server	YES
3	property_id	string	FK → property	YES
4	property_name	string	denorm	
5	buyer_name	string	textField	YES
6	employee_vid	string	session	YES
7	employee_name	string	session	
8	deal_stage	enum	(Initial/Negotiation/OfferSent/ClosedWon/ClosedLost)	YES
9	price_offered	number	numericField	
10	price_final	number	numericField	
11	expected_closing	date	datePicker	YES (sort)
12	actual_closing	date		
13	next_action	enum	DRD	
14	photos	json[]	getImages1	
15	notes	string	3LineBorderForm	
16	gps	json	gpsPosition	
17	status	enum	(OPEN/CLOSED)	YES
18	updated_at	datetime		

Why split 3 tables (not 1)?

Concern	Single table	3 tables
Query "all activity for property X"	Full scan	Index on property_id
Query "open deals"	Filter by status across all rows	Single table scan
Storage cost (Firestore RTDB)	Bigger rows, more nulls	Tighter rows
Update conflicts	Property edit may collide w/ activity insert	Isolated nodes
Index cost	Compound indices needed	Simple per-table
Permissions (Firestore Rules)	Single ruleset across mixed entities	Per-entity rules

Trade-off: need **denorm** of `property_name` ke activity & deal tables (avoid join). Acceptable for read-heavy mobile clients.

13. Bug walkthrough (visual) — Q4 explanation

Lihat row 720 col D (Meeting Owner Kirim button) addToTable:

```

■<5>■5■    ← col 5 table = widget seq#5 value
■<6>■11■    ← col 6 table = widget seq#11 value ← BUG?
■<7>■10■
...
■<11>■4■    ← col 11 = widget #4 (Nama Klien)
■<14>■4■    ← col 14 = widget #4 (Nama Klien) ← DUPLICATE

```

Bug A — col 6 reads separator widget

Widget seq# di Meeting Owner page:

```

seq#1 topMain      (header, no value)
seq#2 separator    (no value)
seq#3 text "Meeting Owner" (static)
seq#4 textField    Nama Klien      ← user input
seq#5 textField    Property        ← user input
seq#6 lqrTextField1 Property QR    ← user input
seq#7 numericField Asking Price    ← user input
seq#8 separator    (no value)
seq#9 DRD          Tipe Properti   ← user input
seq#10 DRD         Lead Source      ← user input
seq#11 separator   (no value)      ← ← ← formula reads INI
seq#12 getImages1  Photo

```

```
seq#13 radio      Hasil
seq#14 textarea  Catatan
seq#15 DRD       Next Action
```

<6>■■■11■ artinya: "isi col 6 dari nilai widget #11". Tapi widget #11 = **separator** (visual spacer, no value). Hasilnya col 6 selalu kosong/null.

Dugaan: should be ■■9■ (Tipe Properti) or ■■12■ (Photo). Need user confirm.

Bug B — col 11 dan col 14 sama-sama Nama Klien

```
<11>■■■4■ Nama Klien → col 11
<14>■■■4■ Nama Klien → col 14
```

Dua kolom DB pegang nilai sama. Possibilities:

1. **Intentional denorm:** col 11 untuk index/search, col 14 untuk display. Common pattern.
2. **Copy-paste error:** salah satu harusnya ref widget lain (e.g. ■■15■ Next Action).

Cara verifikasi: cek detail page (`vertikaTeknoLokaciptaActivityDetail` kalau ada) — kalau col 14 ditampilkan dgn label berbeda dari col 11, berarti bug. Kalau sama-sama "Nama Klien", intentional.

14. Firebase NoSQL constraints

SQL pattern	Firebase RTDB equivalent
WHERE col = val	<code>orderByChild("col").equalTo(val)</code>
WHERE col >= val	<code>orderByChild("col").startAt(val)</code>
ORDER BY col DESC LIMIT N	<code>orderByChild("col").limitToLast(N)</code> (reverse client-side)
JOIN	No JOIN. Denormalize / dual-write / client-side merge
GROUP BY	No GROUP BY. Maintain summary node via write-time fan-out atau dedup di client
Aggregate COUNT/SUM	Counter node maintained on write

Implication for RE:

- Client master table `vtl.realestate-client` is **required** (cannot group-by on activity table)
- Activities denormalize `client_name`, `property_name` (no JOIN at read time)
- Counter cols (e.g. `total_activities` in client master) maintained via Cloud Function trigger atau dual-write pattern (TBD)

15. Final 4-table schema

```
vtl.realestate-client/  ← register klien sekali (master)
vtl.realestate-property/ ← register properti sekali (master, dari Add Listing)
vtl.realestate-activity/ ← log aktivitas (append, denorm client/property)
vtl.realestate-deal/    ← pipeline deal (state machine)
```

vtl.realestate-client cols

Col	Field	Source	Index
1	<code>client_id</code>	NUMBER RE-CL-YYYY-NNNNNN	★
2	<code>created_at</code>	server	★
3	<code>activity_subtype</code>	literal <code>client-add</code>	

Col	Field	Source	Index
4	client_name	textField	★
5	phone	numericField	
6	email	textField	
7	client_type	DRD (Owner/Buyer/Investor/Both)	
8	lead_source	DRD	
9	owned_by_vid	session	★
10	owned_by_name	session	
11	total_activities	counter	
12	status	enum ACTIVE/INACTIVE	★
13	notes	textarea	

vtl.realestate-property cols (Add Listing target)

Col	Field	Source	Index
1	property_id	NUMBER RE - LST - YYYY - NNNNNN	★
2	created_at	server	★
3	created_by_vid	session	★
4	created_by_name	session	
5	property_name	textField	★
6	property_code_qr	lqrTextField (own QR generated)	★
7	location	textField	
8	type_properti	DRD	★
9	listing_type	radio (Sale/Rent)	★
10	asking_price	numericField	
11	negotiable	checkBox	
12	photos	getImages1	
13	gps	auto	
14	status	enum Available/Sold/Hold	★
15	updated_at	server	
16	lead_source	DRD	
17	notes	textarea	

vtl.realestate-activity cols (Meeting/Showing/Follow Up/Marketing/Prospect target)

Col	Field	Source	Index
1	activity_id	NUMBER	★
2	created_at	server	★
3	activity_subtype	literal	★
4	client_id	lqr FK	★
5	client_name	denorm	
6	property_id	lqr FK	★
7	property_name	denorm	
8	property_code_qr	denorm	
9	client_role	literal (owner/buyer)	
10	employee_vid	session	★
11	employee_name	session	
12	cost_center_vid	session	

Col	Field	Source	Index
13	cost_center_name	session	
14	hasil_aktivitas	radio/DRD	★
15	next_action	DRD	
16	photos	getImages	
17	notes	textarea	
18	gps	auto	
21	tipe_properti	DRD nullable	
22	lead_source	DRD nullable	
23	status	literal DONE	
24	asking_price_snapshot	numeric nullable	
25	follow_up_date	datePicker	★
26	channel	nullable (Follow Up only)	
27	conversation_result	nullable (Follow Up only)	
28	marketing_activity	nullable (Marketing only)	

vtl.realestate-deal cols (Deal Progress target)

Col	Field	Source	Index
1	deal_id	NUMBER RE - DEAL - YYYY - NNNNNN	★
2	created_at	server	★
3	property_id	lqr FK	★
4	property_name	denorm	
5	buyer_name	textField (atau FK ke client)	★
6	employee_vid	session	★
7	employee_name	session	
8	deal_stage	DRD	★
9	price_offered	numeric	
10	price_final	numeric	
11	expected_closing	datePicker	★
12	actual_closing	datePicker nullable	
13	next_action	DRD	
14	photos	getImages	
15	notes	textarea	
16	gps	auto	
17	status	enum OPEN/CLOSED	★
18	updated_at	server	

16. Meeting Owner — FINAL addToTable

```

$<env>/<workspace>//vtl.realestate-activity
■retention■4320
■description■report-realestate-activity
■flag■meeting-owner
■tablevid■<RE_ACTIVITY_TABLE_VID>
■index■1★S■3★S■4★S■6★S■10★S■14★S■25★D
■<1>■4■
■<2>■2|T7|yyyy-MM-dd HH:mm:ss■
■<3>■meeting-owner
■<4>■5■
■<5>■5|name■
■<6>■6■
■<7>■6|name■

```

```

■<8>■6|qr■
■<9>■owner
■<10>■session|VID■
■<11>■session|NAME■
■<12>■session|CC_VID■
■<13>■session|CC_NAME■
■<14>■13■
■<15>■16■
■<16>■12■
■<17>■14■
■<18>■2|GPS■
■<21>■9■
■<22>■10■
■<23>■DONE
■<24>■7■
■<25>■15■

```

Widget seq# mapping (post-revisi):

- seq#4 NUMBER → activity_id
- seq#5 lqrTextField client master → client_id + client_name
- seq#6 lqrTextField property master → property_id + property_name + qr
- seq#7 numericField → asking_price snapshot
- seq#9 DRD → tipe_properti
- seq#10 DRD → lead_source
- seq#12 getImages → photos
- seq#13 radio → hasil_aktivitas
- seq#14 textarea → notes
- seq#15 datePicker → follow_up_date
- seq#16 DRD → next_action

17. Add Client form (NEW page — prerequisite master)

Route key: vertikaTeknoLokaciptaAddClient

Page widgets (op1Screen rows TBD)

```

seq#1  topMain
seq#2  separator
seq#3  text "Tambah Klien Baru"
seq#4  NUMBER          template: RE-CL-{{YYYY}}-{{COUNTER(vtl.realestate-client,6)}}
      position 4
seq#5  textField        Nama Lengkap          position 5
seq#6  numericField     No. HP                position 6
seq#7  textField        Email                 position 7
seq#8  displayDropDownData Tipe Klien          option: [Owner, Buyer, Investor, Both] position 8
seq#9  displayDropDownData Lead Source          option: [Canvassing, Referral, Online, Walk-in, Social Media, Cold-call...]
seq#10 separator
seq#11 3LineBorderForm  Catatan                position 11
seq#12 sendButtonGpsAddTable position 251

```

addToTable

```

$<env>/<workspace>/vtl.realestate-client
■retention■4320
■description■client-registration
■flag■client-add
■tablevid■<CLIENT_TABLE_VID>
■index■1★S■2★D■4★S■9★S■12★S
■<1>■4■
■<2>■2|T7|yyyy-MM-dd HH:mm:ss■
■<3>■client-add

```

```

■<4>■5■
■<5>■6■
■<6>■7■
■<7>■8■
■<8>■9■
■<9>■session|VID■
■<10>■session|NAME■
■<11>■0■
■<12>■ACTIVE
■<13>■11■

```

Generator run: "run":

```

"5:disable◆6:disable◆7:disable◆8:disable◆9:disable◆11:disable◆4:generate_number◆251:disable"

```

18. Add Listing form refactor (existing row 722–741)

Patch widget seq# layout (insert NUMBER, switch property text → property generates own QR)

```

seq#1 topMain
seq#2 separator
seq#3 text "Add Listing"
seq#4 NUMBER          template: RE-LST-{{YYYY}}-{{COUNTER(vtl.realestate-property,6)}} position 4
seq#5 textField       Nama Properti          position 5
seq#6 textField       Lokasi                  position 6
seq#7 displayDropDownData Tipe Properti      position 7
seq#8 radio           Listing Type (Sale/Rent) position 8
seq#9 numericField    Asking Price            position 9
seq#10 checkBox       Negotiable              position 10
seq#11 displayDropDownData Lead Source        position 11
seq#12 getImages1     Property Photo          position 12
seq#13 3LineBorderForm Catatan                position 13
seq#14 sendButtonGpsAddTable position 251

```

addToTable

```

$<env>/<workspace>//vtl.realestate-property
■retention■4320
■description■property-listing
■flag■property-add
■tablevid■<PROPERTY_TABLE_VID>
■index■1★S■2★D■3★S■5★S■6★S■8★S■9★S■14★S
■<1>■4■
■<2>■2|T7|yyyy-MM-dd HH:mm:ss■
■<3>■session|VID■
■<4>■session|NAME■
■<5>■5■
■<6>■4■
■<7>■6■
■<8>■7■
■<9>■8■
■<10>■9■
■<11>■10■
■<12>■12■
■<13>■2|GPS■
■<14>■Available
■<15>■2|T7|yyyy-MM-dd HH:mm:ss■
■<16>■11■
■<17>■13■

```

<6> property_code_qr = pakai property_id itu sendiri (col 1) sebagai QR payload — agent scan QR di lapangan → langsung dapat ID.

19. Meeting Buyer form refactor (existing row 762–774)

```

seq#1 topMain
seq#2 separator
seq#3 text "Meeting Buyer"
seq#4 NUMBER          template: RE-MTG-{{YYYY}}-{{COUNTER(vtl.realestate-activity,6)}} position 4
seq#5 lqrTextField    Pilih Klien (buyer) table=vtl.realestate-client position 5
seq#6 lqrTextField1   Pilih Properti    table=vtl.realestate-property position 6
seq#7 displayDropDownData Hasil Aktivitas [Potential, Follow Up, Not Interested] position 7
seq#8 getImages1      Photo              position 8
seq#9 3LineBorderForm Catatan            position 9
seq#10 TXF date        Jadwal Follow-up  position 10
seq#11 displayDropDownData Next Action    position 11
seq#12 sendButtonGpsAddTable position 251

```

addToTable

```

$<env>/<workspace>/vtl.realestate-activity
■retention■4320
■description■report-realestate-activity
■flag■meeting-buyer
■tablevid■<RE_ACTIVITY_TABLE_VID>
■index■1★S■3★S■4★S■6★S■10★S■14★S■25★D
■<1>■4■
■<2>■2|T7|yyyy-MM-dd HH:mm:ss■
■<3>■meeting-buyer
■<4>■5■
■<5>■5|name■
■<6>■6■
■<7>■6|name■
■<8>■6|qr■
■<9>■buyer
■<10>■session|VID■
■<11>■session|NAME■
■<12>■session|CC_VID■
■<13>■session|CC_NAME■
■<14>■7■
■<15>■11■
■<16>■8■
■<17>■9■
■<18>■2|GPS■
■<23>■DONE
■<25>■10■

```

20. Showing form refactor (existing row 775–787)

```

seq#1 topMain
seq#2 separator
seq#3 text "Showing"
seq#4 NUMBER          template: RE-SHW-{{YYYY}}-{{COUNTER(vtl.realestate-activity,6)}}
seq#5 lqrTextField    Pilih Buyer        position 5
seq#6 lqrTextField1   Pilih Properti      position 6
seq#7 displayDropDownData Hasil Jenis Interest [High, Medium, Low] position 7
seq#8 getImages1      Photo              position 8
seq#9 3LineBorderForm Catatan            position 9
seq#10 TXF date        Jadwal Follow-up  position 10
seq#11 displayDropDownData Next Action    position 11
seq#12 sendButtonGpsAddTable

```

addToTable

```

$<env>/<workspace>/vtl.realestate-activity
■retention■4320■description■report-realestate-activity■flag■showing
■tablevid■<RE_ACTIVITY_TABLE_VID>
■index■1★S■3★S■4★S■6★S■10★S■14★S■25★D
■<1>■4■■<2>■2|T7|yyyy-MM-dd HH:mm:ss■■<3>■showing
■<4>■5■■<5>■5|name■
■<6>■6■■<7>■6|name■■<8>■6|qr■
■<9>■buyer
■<10>■session|VID■■<11>■session|NAME■■<12>■session|CC_VID■■<13>■session|CC_NAME■
■<14>■7■■<15>■11■
■<16>■8■■<17>■9■■<18>■2|GPS■
■<23>■DONE■<25>■10■

```

21. Follow Up form refactor (existing row 788–802)

```
seq#1 topMain
seq#2 separator
seq#3 text "Follow Up" ← FIX title (was "Showing")
seq#4 NUMBER RE-FUP-{{YYYY}}-{{COUNTER(vtl.realestate-activity,6)}}
seq#5 lqrTextField Pilih Klien
seq#6 lqrTextField1 Pilih Properti
seq#7 displayDropDownData Channel [WhatsApp, Telp, Email, Direct] position 7
seq#8 displayDropDownData Conversation Result [Reached, Voicemail, No Answer] position 8
seq#9 getImages1 Photo
seq#10 3LineBorderForm Catatan
seq#11 TXF date Jadwal Follow-up berikutnya
seq#12 displayDropDownData Next Action
seq#13 sendButtonGpsAddTable
```

addToTable

```
$(env)/<workspace>/vtl.realestate-activity
■retention■4320■description■report-realestate-activity■flag■follow-up
■tablevid■<RE_ACTIVITY_TABLE_VID>
■index■1★S■3★S■4★S■6★S■10★S■14★S■25★D
■<1>■4■<2>■2|T7|yyy-MM-dd HH:mm:ss■<3>■follow-up
■<4>■5■<5>■5|name■
■<6>■6■<7>■6|name■<8>■6|qr■
■<9>■buyer
■<10>■session|VID■<11>■session|NAME■<12>■session|CC_VID■<13>■session|CC_NAME■
■<14>■8■<15>■12■
■<16>■9■<17>■10■<18>■2|GPS■
■<23>■DONE■<25>■11■
■<26>■7■<27>■8■
```

22. Marketing Property form refactor (existing row 742–761)

```
seq#1 topMain
seq#2 separator
seq#3 text "Marketing Property"
seq#4 NUMBER RE-MKT-{{YYYY}}-{{COUNTER(vtl.realestate-activity,6)}}
seq#5 lqrTextField1 Pilih Properti
seq#6 radio Marketing Activity [Brochure, Open House, Tour, Ad]
seq#7 numericField Reach / Impressions
seq#8 getImages1 Photo
seq#9 3LineBorderForm Catatan
seq#10 TXF date Jadwal Follow-up
seq#11 displayDropDownData Next Action
seq#12 sendButtonGpsAddTable
```

addToTable

```
$(env)/<workspace>/vtl.realestate-activity
■retention■4320■description■report-realestate-activity■flag■marketing
■tablevid■<RE_ACTIVITY_TABLE_VID>
■index■1★S■3★S■6★S■10★S■25★D
■<1>■4■<2>■2|T7|yyy-MM-dd HH:mm:ss■<3>■marketing
■<6>■5■<7>■5|name■<8>■5|qr■
■<10>■session|VID■<11>■session|NAME■<12>■session|CC_VID■<13>■session|CC_NAME■
■<15>■11■
■<16>■8■<17>■9■<18>■2|GPS■
■<23>■DONE■<25>■10■
■<28>■6■
```

(Marketing tidak punya client — col 4,5,9,14 NULL)

23. Prospect Owner form refactor (existing row 820–836+)

```
seq#1 topMain
seq#2 separator
seq#3 text "Prospect Owner"
seq#4 NUMBER RE-PRS-{{YYYY}}-{{COUNTER(vtl.realestate-activity,6)}}
seq#5 textField Nama Klien (jika belum di-master, save langsung)
seq#6 lqrTextField1 Pilih Properti (existing) OR text Lokasi (new prospect)
seq#7 displayDropDownData Tipe Properti
seq#8 displayDropDownData Lead Source
seq#9 getImages1 Photo
seq#10 radio Hasil Aktivitas
seq#11 3LineBorderForm Keterangan
seq#12 TXF date Jadwal Follow-up
seq#13 displayDropDownData Next Action
seq#14 sendButtonGpsAddTable
```

addToTable

```
$<env>/<workspace>//vtl.realestate-activity
■retention■4320■description■report-realestate-activity■flag■prospect-owner
■tablevid■<RE_ACTIVITY_TABLE_VID>
■index■1★S■3★S■4★S■6★S■10★S■14★S■25★D
■<1>■4■<2>■2|T7|yyyy-MM-dd HH:mm:ss■<3>■prospect-owner
■<5>■5■
■<6>■6■<7>■6|name■<8>■6|qr■
■<9>■owner
■<10>■session|VID■<11>■session|NAME■<12>■session|CC_VID■<13>■session|CC_NAME■
■<14>■10■<15>■13■
■<16>■9■<17>■11■<18>■2|GPS■
■<21>■7■<22>■8■
■<23>■DONE■<25>■12■
```

(Prospect Owner: client belum di-master → simpan nama as text col 5, client_id col 4 NULL → backend cleanup later)

24. Deal Progress form refactor (existing row 803–819)

Target table: `vtl.realestate-deal` (beda dari activity)

```
seq#1 topMain
seq#2 separator
seq#3 text "Deal Progress"
seq#4 NUMBER RE-DEAL-{{YYYY}}-{{COUNTER(vtl.realestate-deal,6)}} position 4
seq#5 lqrTextField Pilih Buyer (client master) position 5
seq#6 lqrTextField1 Pilih Properti position 6
seq#7 displayDropDownData Deal Stage [Initial, Negotiation, OfferSent, ClosedWon, ClosedLost] position 7
seq#8 numericField Price Offered position 8
seq#9 numericField Price Final position 9
seq#10 TXF date Expected Closing position 10
seq#11 getImages1 Photo position 11
seq#12 3LineBorderForm Catatan position 12
seq#13 displayDropDownData Next Action position 13
seq#14 sendButtonGpsAddTable position 251
```

addToTable

```
$<env>/<workspace>//vtl.realestate-deal
■retention■4320■description■deal-pipeline■flag■deal-create
■tablevid■<DEAL_TABLE_VID>
■index■1★S■2★D■3★S■5★S■6★S■8★S■11★D■17★S
■<1>■4■
■<2>■2|T7|yyyy-MM-dd HH:mm:ss■
■<3>■6■<4>■6|name■
■<5>■5|name■
■<6>■session|VID■<7>■session|NAME■
■<8>■7■
■<9>■8■<10>■9■
■<11>■10■
```

```

■<13>■13■
■<14>■11■<15>■12■<16>■2|GPS■
■<17>■OPEN
■<18>■2|T7|yyyy-MM-dd HH:mm:ss■

```

Deal Stage update (updateTableRow — agent edit existing deal)

Untuk edit deal yg sudah ada (e.g. stage berubah dari Negotiation → OfferSent):

```

$<env>/<workspace>//vtl.realestate-deal
■action■update
■tablevid■<DEAL_TABLE_VID>
■search■1★<deal_id>
■<8>■<new deal_stage>
■<10>■<new price_final>
■<11>■<new expected_closing>
■<13>■<new next_action>
■<17>■<new status: OPEN or CLOSED>
■<18>■2|T7|yyyy-MM-dd HH:mm:ss■

```

Trigger via button di Deal Detail page. Tombol "Update Status".

25. Activity Detail page (NEW — universal untuk semua activity subtype)

Route key: vertikaTeknoLokaciptaActivityDetail

Param: activity_id

```

{
  "title": "Vertika Tekno Lokacipta",
  "children": [
    {
      "type": "HGR", "width": 86, "row": 1, "fontSize": 13, "beforeSpacing": 5, "afterSpacing": 0, "sortBy": "alphabet...",
      {
        "url": "<autsorz-icon>", "text": "Autsorz", "route": "autsorz"},
        {
        "url": "<vtl-icon>", "text": "VTL", "route": "vertikaTeknoLokacipta"},
        {
        "url": "<info-icon>", "text": "Info", "route": "info"}
      }
    },
    {
      "type": "TXT", "afterSpacing": 0, "size": 4, "data": " ",
      {
        "type": "WORKER_CARD_DETAIL",
        "role": "AGENT",
        "vidtable": "<RE_ACTIVITY_TABLE_VID>",
        "table": "$<env>/<workspace>//vtl.realestate-activity",
        "search": "1■<activity_id>",
        "conditions": "[[■1■<activity_id>]]",
        "text": "AGENT◆<11>◆<3> . <14>"
      }
    },
    {
      "type": "displayStaticField",
      "table": "$<env>/<workspace>//vtl.realestate-activity",
      "search": "1■<activity_id>",
      "fields": [
        {
        "label": "Activity ID", "value": "<1>"},
        {
        "label": "Type", "value": "<3>"},
        {
        "label": "Created", "value": "<2>"},
        {
        "label": "Client", "value": "<5>", "route": "vertikaTeknoLokaciptaClientDetail■client_id■<4>"},
        {
        "label": "Property", "value": "<7>", "route": "vertikaTeknoLokaciptaPropertyDetail■property_id■<6>"},
        {
        "label": "Asking Price", "value": "<24>"},
        {
        "label": "Tipe Properti", "value": "<21>"},
        {
        "label": "Lead Source", "value": "<22>"},
        {
        "label": "Hasil", "value": "<14>"},
        {
        "label": "Next Action", "value": "<15>"},
        {
        "label": "Follow-up", "value": "<25>"},
        {
        "label": "Channel", "value": "<26>"},
        {
        "label": "Conv. Result", "value": "<27>"},
        {
        "label": "Catatan", "value": "<17>"
      }
    ]
  }
},
{
  "type": "photoDisplay2",

```

```

"table": "$<env>/<workspace>/vtl.realestate-activity",
"search": "1■<activity_id>",
"image": "<16>"
},
{
  "type": "COMMENT_SECTION",
  "role": "AGENT",
  "ledgerCode": "RE-ACTIVITY-CONVERSATION",
  "vidtable": "<RE_COMMENT_TABLE_VID>",
  "table": "$<env>/<workspace>/vtl.realestate-comment",
  "search": "2■<activity_id>",
  "text": "DISKUSI",
  "content": "<3>◆<4>◆<5>◆<6>"
},
{
  "type": "txf",
  "variant": "commentBox",
  "role": "AGENT",
  "vidtable": "<RE_COMMENT_TABLE_VID>",
  "table": "$<env>/<workspace>/vtl.realestate-comment",
  "text": "Tambah komentar...◆Kirim◆Lampiran◆Komentar kosong",
  "icon": "attach_file",
  "label": "",
  "hint": "Tambah komentar...",
  "maxLength": 500,
  "position": 7,
  "buttonIcon": "send",
  "disabledWhenEmpty": true,
  "attachmentEnabled": true,
  "attachmentFolder": "id/<workspace>/vtl/realestate-comment/<activity_id>",
  "attachmentFilename": "<activity_id>-<timestamp>",
  "attachmentMax": 3,
  "addToTable": "$<env>/<workspace>/vtl.realestate-comment■retention■4320■<2>■<activity_id>■<3>■AGENT■<4>■sessi
}
]
}

```

26. Client Detail page (NEW)

Route: `vertikaTeknoLokaciptaClientDetail`

Param: `client_id`

```

{
  "title": "Detail Klien",
  "children": [
    { "type": "HGR", "...": "<header>" },
    {
      "type": "displayStaticField",
      "table": "$<env>/<workspace>/vtl.realestate-client",
      "search": "1■<client_id>",
      "fields": [
        { "label": "Client ID", "value": "<1>" },
        { "label": "Nama", "value": "<4>" },
        { "label": "HP", "value": "<5>" },
        { "label": "Email", "value": "<6>" },
        { "label": "Tipe", "value": "<7>" },
        { "label": "Lead Source", "value": "<8>" },
        { "label": "Owner Agent", "value": "<10>" },
        { "label": "Total Aktivitas", "value": "<11>" },
        { "label": "Status", "value": "<12>" },
        { "label": "Catatan", "value": "<13>" }
      ]
    },
    { "type": "TXT", "size": 18, "data": "Riwayat Aktivitas", "margin": "16,0,0,16" },
    {
      "type": "displayTableInteractive",
      "table": "$<env>/<workspace>/vtl.realestate-activity",
      "vidtable": "<RE_ACTIVITY_TABLE_VID>",
      "search": "4■<client_id>",
      "conditions": "[[■4■<client_id>]]",
      "orderByChild": "2",
      "limitToLast": 50,
      "sortDesc": true,
      "text": "Aktivitas◆<3>◆<7>◆<14>◆<25>◆<2>"
    }
  ]
}

```

```

    "route": "vertikaTeknoLokaciptaActivityDetail■activity_id■<1>"
  }
]
}

```

27. Property Detail page (NEW)

Route: `vertikaTeknoLokaciptaPropertyDetail`

Param: `property_id`

```

{
  "title": "Detail Properti",
  "children": [
    {
      "type": "HGR", "...": "<header>",
      {
        "type": "displayStaticField",
        "table": "$<env>/<workspace>//vtl.realestate-property",
        "search": "1■<property_id>",
        "fields": [
          {"label": "Property ID", "value": "<1>"},
          {"label": "Nama", "value": "<5>"},
          {"label": "QR", "value": "<6>"},
          {"label": "Lokasi", "value": "<7>"},
          {"label": "Tipe", "value": "<8>"},
          {"label": "Listing", "value": "<9>"},
          {"label": "Asking Price", "value": "<10>"},
          {"label": "Negotiable", "value": "<11>"},
          {"label": "Lead Source", "value": "<16>"},
          {"label": "Status", "value": "<14>"}
        ]
      },
    {
      "type": "photoDisplay2",
      "table": "$<env>/<workspace>//vtl.realestate-property",
      "search": "1■<property_id>",
      "image": "<12>"
    },
    {"type": "TXT", "size": 18, "data": "Riwayat Aktivitas Properti"},
    {
      "type": "displayTableInteractive",
      "table": "$<env>/<workspace>//vtl.realestate-activity",
      "search": "6■<property_id>",
      "conditions": "[[6■<property_id>]]",
      "orderByChild": "2",
      "limitToLast": 50,
      "sortDesc": true,
      "text": "Aktivitas◆<3>◆<5>◆<14>◆<2>",
      "route": "vertikaTeknoLokaciptaActivityDetail■activity_id■<1>"
    },
    {"type": "TXT", "size": 18, "data": "Deal Pipeline"},
    {
      "type": "displayTableInteractive",
      "table": "$<env>/<workspace>//vtl.realestate-deal",
      "search": "3■<property_id>",
      "conditions": "[[3■<property_id>]]",
      "orderByChild": "2",
      "limitToLast": 20,
      "sortDesc": true,
      "text": "Deal◆<8>◆<5>◆<10>◆<11>",
      "route": "vertikaTeknoLokaciptaDealDetail■deal_id■<1>"
    }
  ]
}

```

28. Deal Detail page (NEW — with status update buttons)

Route: `vertikaTeknoLokaciptaDealDetail`

Param: deal_id

```
{
  "title": "Detail Deal",
  "children": [
    {
      "type": "HGR", "...": "<header>",
      {
        "type": "displayStaticField",
        "table": "$<env>/<workspace>//vtl.realestate-deal",
        "search": "1■<deal_id>",
        "fields": [
          {"label": "Deal ID", "value": "<1>"},
          {"label": "Property", "value": "<4>", "route": "vertikaTeknoLokaciptaPropertyDetail■property_id■<3>"},
          {"label": "Buyer", "value": "<5>"},
          {"label": "Agent", "value": "<7>"},
          {"label": "Stage", "value": "<8>"},
          {"label": "Price Offered", "value": "<9>"},
          {"label": "Price Final", "value": "<10>"},
          {"label": "Expected Closing", "value": "<11>"},
          {"label": "Actual Closing", "value": "<12>"},
          {"label": "Next Action", "value": "<13>"},
          {"label": "Catatan", "value": "<15>"},
          {"label": "Status", "value": "<17>"}
        ]
      },
    ],
    {
      "type": "RBT",
      "alignment": "spaceEvenly",
      "children": [
        {
          "text": "Next Stage",
          "color": "blue",
          "icon": "arrow_forward",
          "route": "vertikaTeknoLokaciptaDealUpdate■deal_id■<deal_id>",
          "delay": 5
        },
        {
          "text": "Tutup Deal",
          "color": "green",
          "icon": "check",
          "updateTableRow": "$<env>/<workspace>//vtl.realestate-deal■tablevid■<DEAL_TABLE_VID>■search■1★<deal_id>■<8>.",
          "chain": {
            "type": "DO_DIALOG",
            "title": "Deal Closed",
            "children": [
              {
                "type": "TXT", "data": "Selamat! Deal ditutup."},
              {
                "type": "RBT", "alignment": "center", "children": [{"text": "OK", "route": "vertikaTeknoLokaciptaMyDea..."}]
              }
            ]
          }
        },
        {
          "text": "Batal Deal",
          "color": "red",
          "icon": "close",
          "updateTableRow": "$<env>/<workspace>//vtl.realestate-deal■tablevid■<DEAL_TABLE_VID>■search■1★<deal_id>■<8>.",
          "chain": {
            "type": "DO_DIALOG",
            "title": "Deal Batal",
            "children": [
              {
                "type": "TXT", "data": "Deal ditandai batal."},
              {
                "type": "RBT", "alignment": "center", "children": [{"text": "OK", "route": "vertikaTeknoLokaciptaMyDea..."}]
              }
            ]
          }
        }
      ]
    }
  ]
}
```

29. List "Klien Saya"

Route: vertikaTeknoLokaciptaMyClients

```
{
  "title": "Klien Saya",
```

```

"children": [
  {"type": "HGR", "...": "<header>"},
  {"type": "TXT", "size": 24, "data": "Klien Saya"},
  {"type": "TXT", "afterSpacing": 0, "size": 4, "data": " "},
  {
    "type": "RBT",
    "alignment": "end",
    "children": [{
      "text": "+ Tambah Klien",
      "color": "primary",
      "route": "vertikaTeknoLokaciptaAddClient",
      "delay": 5
    }]
  },
  {
    "type": "displayTableInteractive",
    "table": "$<env>/<workspace>/vtl.realestate-client",
    "vidtable": "<CLIENT_TABLE_VID>",
    "search": "9█<session.vid>",
    "conditions": "[[█9█<session.vid>],[█12█ACTIVE]]",
    "orderByChild": "2",
    "limitToLast": 100,
    "sortDesc": true,
    "text": "Klien<4>◆<7>◆<5>◆<11> aktivitas◆<2>",
    "route": "vertikaTeknoLokaciptaClientDetail█client_id█<1>"
  }
]
}

```

30. List "Pipeline Deal"

Route: `vertikaTeknoLokaciptaMyDeals`

```

{
  "title": "Pipeline Deal",
  "children": [
    {"type": "HGR", "...": "<header>"},
    {"type": "TXT", "size": 24, "data": "Pipeline Saya"},
    {"type": "TXT", "afterSpacing": 0, "size": 4, "data": " "},
    {
      "type": "displayTableInteractive",
      "table": "$<env>/<workspace>/vtl.realestate-deal",
      "vidtable": "<DEAL_TABLE_VID>",
      "search": "6█<session.vid>",
      "conditions": "[[█6█<session.vid>],[█17█OPEN]]",
      "orderByChild": "11",
      "limitToLast": 50,
      "text": "Deal◆<8>◆<4>◆<5>◆<10>◆Closing: <11>",
      "route": "vertikaTeknoLokaciptaDealDetail█deal_id█<1>"
    }
  ]
}

```

31. List "Today Follow Up"

Route: `vertikaTeknoLokaciptaTodayFollowUp`

```

{
  "title": "Follow-up Hari Ini",
  "children": [
    {"type": "HGR", "...": "<header>"},
    {"type": "TXT", "size": 24, "data": "Follow-up Hari Ini"},
    {
      "type": "displayTableInteractive",
      "table": "$<env>/<workspace>/vtl.realestate-activity",
      "vidtable": "<RE_ACTIVITY_TABLE_VID>",
      "search": "10█<session.vid>",
      "conditions": "[[█10█<session.vid>],[█25█<today>█],[█23█DONE]]",
      "orderByChild": "25",
    }
  ]
}

```

```

    "limitToLast": 50,
    "sortDesc": false,
    "text": "Follow-up♦<5>♦<7>♦<14>♦<25>♦<15>",
    "route": "vertikaTeknoLokaciptaActivityDetail■activity_id■<1>"
  }
}
}

```

Note: <today> placeholder resolves client-side ke ISO date hari ini. conditions filter follow_up_date <= today butuh range query Firebase: orderByChild("25").endAt("<today>"). DSL syntax tepat TBD — verify dgn existing pending-approval-page.json pattern.

32. List "Aktivitas Terbaru"

Route: vertikaTeknoLokaciptaRecentActivity

```

{
  "title": "Aktivitas Terbaru",
  "children": [
    { "type": "HGR", "...": "<header>" },
    { "type": "TXT", "size": 24, "data": "Aktivitas 30 Hari Terakhir"},
    {
      "type": "displayTableInteractive",
      "table": "$<env>/<workspace>/vtl.realestate-activity",
      "vidtable": "<RE_ACTIVITY_TABLE_VID>",
      "search": "10■<session.vid>",
      "conditions": "[[■10■<session.vid>]]",
      "orderByChild": "2",
      "limitToLast": 100,
      "sortDesc": true,
      "text": "Aktivitas♦<3>♦<5>♦<7>♦<14>♦<2>",
      "route": "vertikaTeknoLokaciptaActivityDetail■activity_id■<1>"
    }
  ]
}

```

33. Vertika home menu — tambah entry list

Route handler vertikaTeknoLokacipta (home page) tambah menu items:

+ Tambah Klien	→ vertikaTeknoLokaciptaAddClient
Klien Saya	→ vertikaTeknoLokaciptaMyClients
Pipeline Deal	→ vertikaTeknoLokaciptaMyDeals
Follow-up Hari Ini	→ vertikaTeknoLokaciptaTodayFollowUp
Aktivitas Terbaru	→ vertikaTeknoLokaciptaRecentActivity
Lapor Aktivitas:	
Meeting Owner	→ vertikaTeknoLokaciptaReportMeetingOwner
Meeting Buyer	→ vertikaTeknoLokaciptaReportMeetingBuyer
Showing	→ vertikaTeknoLokaciptaReportShowing
Follow Up	→ vertikaTeknoLokaciptaReportFollowUp
Marketing	→ vertikaTeknoLokaciptaReportMarketing
Prospect Owner	→ vertikaTeknoLokaciptaReportProspect
Add Listing	→ vertikaTeknoLokaciptaReportAddListing
Deal Progress	→ vertikaTeknoLokaciptaReportDealProgress

34. Firebase Rules sketch

```

{
  "rules": {
    "$env": {
      "$workspace": {
        "vtl_realestate-client": {

```

```

        ".indexOn": ["1", "2", "4", "9", "12"],
        "$row": {
            ".read": "auth.uid != null && data.child('9').val() == auth.uid",
            ".write": "auth.uid != null && newData.child('9').val() == auth.uid"
        }
    },
    "vtl_realestate-property": {
        ".indexOn": ["1", "2", "3", "5", "6", "8", "9", "14"],
        ".read": "auth.uid != null",
        "$row": {
            ".write": "auth.uid != null && (!data.exists() || data.child('3').val() == auth.uid)"
        }
    },
    "vtl_realestate-activity": {
        ".indexOn": ["1", "2", "3", "4", "6", "10", "14", "25"],
        "$row": {
            ".read": "auth.uid != null && data.child('10').val() == auth.uid",
            ".write": "auth.uid != null && newData.child('10').val() == auth.uid && (!data.exists() || data.child('10'..."
        }
    },
    "vtl_realestate-deal": {
        ".indexOn": ["1", "2", "3", "5", "6", "8", "11", "17"],
        "$row": {
            ".read": "auth.uid != null && data.child('6').val() == auth.uid",
            ".write": "auth.uid != null && newData.child('6').val() == auth.uid"
        }
    },
    "vtl_realestate-comment": {
        ".indexOn": ["1", "2", "4"],
        "$row": {
            ".read": "auth.uid != null",
            ".write": "auth.uid != null && newData.child('4').val() == auth.uid"
        }
    }
}
}
}
}
}
}

```

Property = world-readable in workspace (semua agent boleh lihat listing apa saja). Client + Activity + Deal = scoped ke owner agent. Comment = readable all, writable own.

35. Build order checklist

- 1. Create Firebase nodes:
vtl.realstate-client / property / activity / deal / comment
- 2. Apply Firebase Rules (section 34)
- 3. Build Add Client form (section 17) → seed master
- 4. Build Add Listing form (section 18 refactor) → seed property master
- 5. Build Activity Detail page (section 25) → universal detail
- 6. Build Client Detail page (section 26)
- 7. Build Property Detail page (section 27)
- 8. Build Deal Detail page (section 28)
- 9. Refactor 6 activity forms (sections 19-23, Meeting Owner section 16)
- 10. Build Deal Progress form (section 24)
- 11. Build 4 list pages (sections 29-32)
- 12. Update Vertika home menu (section 33)
- 13. End-to-end smoke test:
Add Client → Add Listing → Meeting Owner → Follow Up → Deal Progress → Close Deal
Verify each row appears in correct list, detail pages load, status updates persist.

36. Cross-page wiring (route → detail flow)

Existing pattern (pending-approval-page.json line 12):

```
"route": "vertikaTeknoLokaciptaRequestDetail■no_request■<1>"
```

Saat user tap row di list, navigate ke detail page dgn param `no_request = col1 value (= request ID)`. Detail page baca param via `<no_request>` placeholder, lookup row di Firebase via `search: "1■<no_request>".`

Untuk Real Estate, list page route should be:

```
vertikaTeknoLokaciptaRealEstateActivityDetail■activity_id■<1>
```

Detail page baca `<activity_id>`, query `vtl.realstate-activity` where `col1 = activity_id`, render fields.

Approve/Reject flow (kalau perlu approval) pakai `updateTableRow` w/ `search■1★<activity_id>` patch col status.